



Testing Evidence and Limitations

What has actually been proven to work, how it was proven, and everything that has not

1. The headline result

A real person sent an Arabic voice note over WhatsApp and received a professional bilingual PDF CV back, with no human intervention at any point in the chain.

That is the whole product thesis, and it is validated rather than asserted.

2. The end-to-end run

Conducted on 18 July 2026 with a real user on a real phone.

Stage	Result
Voice note received via Twilio	Media downloaded successfully
Groq Whisper transcription	Jordanian dialect transcribed accurately
Interview	12 messages, one question at a time, warm colloquial Arabic throughout
Extraction	Complete structured profile built incrementally across turns
CV content	Claude produced parallel English and Modern Standard Arabic
PDF rendering	Gotenberg rendered correct Arabic shaping and RTL layout
Storage	Uploaded to the <code>cvs</code> bucket, path written to <code>profiles.cv_pdf_path</code>
Delivery	152 KB PDF received in the WhatsApp thread
Final state	<code>state = cv_sent</code>

The profile extracted: كرم شهودقة — private driver — three employers, approximately five years each — defensive driving, passenger safety, route planning, vehicle maintenance — and an achievement about stopping an attempted car theft on a desert road and getting the family home safely.

What the CV writer did with it: correctly inferred *fifteen years of experience across three companies* from three five-year entries — an inference from the data, not an invention — and led the summary with that

figure. The achievement was written up in professional language in both languages. **Nothing in the CV was fabricated; every fact traces to something the worker said.**

The user confirmed during the interview that the agent was "understanding me correctly."

3. Component-level verification

Interview quality. Verified in live use: one question per message, dialect maintained, prior answers acknowledged, no repeated questions, and completion triggered at the right moment rather than mid-conversation.

Arabic PDF rendering. The highest-risk component, verified early and separately via `cv/preview/render_preview.py`, which renders the template locally with sample data. This allowed layout and typography to be iterated in seconds instead of through WhatsApp round trips, and confirmed correct letter shaping before any of it was wired up.

Privacy controls. Actively tested, not assumed. The anonymous role was queried directly and confirmed **unable** to read `full_name`, `users`, `conversations` or `matches`. Supabase's advisor was run and its `security_definer_view` finding was fixed and re-verified.

Front-end surfaces. Dashboard, landing page and employer portal were each loaded and confirmed to render live data, in both Arabic and English, including CV links and the trade chart.

Architecture diagram. Reviewed visually and spell-checked.

Documentation. Every PDF in this set was rendered and a sample inspected visually.

4. Bugs found by testing, and how

Each of these was found by a real person using the product or by the self-test endpoint — none by reading code.

Bug	Symptom	Found by	Cause
<code>temperature</code> rejected	Silent pipeline failure	<code>&claude=1</code>	Claude Sonnet rejects the parameter
Twilio 63007	No outbound messages	<code>&sendto=</code>	<code>TWILIO_WHATSAPP_FROM</code> not WhatsApp-enabled
Gotenberg unreachable	No CV delivered	<code>&cvtest=</code>	URL configured without <code>https://</code>
ElevenLabs 402	Text replies only	<code>&ttstest=1</code>	Voice library needs a paid plan
<code>security_definer_view</code>	—	Supabase advisor	Views missing <code>security_invoker = true</code>
Dashboard "failed to fetch"	Demo data shown	User report	Key and view configuration

Bug	Symptom	Found by	Cause
No CV link on dashboard	User report	User report	<code>cv_pdf_path</code> not exposed in the view
Twilio 63038 daily cap	Replies generated, never delivered	User report → <code>&sendto=</code>	Trial account limited to 50 outbound messages/day

The pattern is worth stating: **the self-test endpoint found four production bugs in minutes each, after hours had been spent guessing at a black box**. Building observability was the highest-return decision of the project.

The silent-failure class, and the fix

The Twilio 63038 case exposed something worse than the rate limit itself. Outbound replies were written to `conversations` **before** the send was attempted, and send errors were caught and logged to a console stream Supabase does not expose. The database, the dashboard and the request logs therefore all looked perfectly healthy while **nothing was reaching users**.

Migration `0005_delivery_status.sql` closes this: an outbound turn is written as `pending`, the send result is captured, and the row is resolved to `sent` (with the Twilio SID) or `failed` (with the provider's error text). `generate-cv` no longer advances a user to `cv_sent` unless Twilio actually accepted the message, and the self-test endpoint surfaces recent failures directly.

The general lesson, and the one worth carrying forward: **never record an outcome before the thing that determines it has happened**. A success written optimistically is indistinguishable from a real success, which turns a loud failure into a silent one.

5. What has *not* been tested

Stated plainly, because an untested claim is not evidence.

- **Scale.** The system has handled a handful of conversations. It has never seen concurrent users, and there is no load testing, rate limiting, or queueing.
- **Dialect breadth.** Validated on Jordanian/Levantine Arabic. Gulf, Egyptian and Maghrebi dialects are untested, and Maghrebi in particular is likely to be the hardest for transcription.
- **Poor audio conditions.** All test recordings were reasonably clear. Street noise, workshops, vehicle interiors and low-end microphones — the actual conditions of the target audience — are untested.
- **Long and rambling voice notes.** Only short, focused answers have been exercised. History is truncated to the last ten turns, so very long conversations may lose early context.
- **Adversarial input.** Prompt-injection defences are implemented and reasoned about, but have not been red-teamed.
- **Interrupted conversations.** A user who abandons the interview and returns days later has not been tested.

- **The employer side end to end.** Browsing and CV links work; no real employer has used the portal to hire anyone.
- **Placement recording.** The table and counter exist; no real placement has been recorded.
- **Cross-device WhatsApp rendering.** The PDF has been confirmed delivered, but not visually inspected across a range of Android and iOS WhatsApp PDF viewers.
- **Automated tests.** There is no test suite. All verification was manual. This is an explicit hackathon trade-off, and it is the first thing to add if the project continues.

6. Functional limitations

- **Voice replies are disabled.** Built and deployed behind a flag; blocked by the ElevenLabs free plan. Until it is enabled, the agent's side of the conversation still requires reading, which leaves the literacy loop only half closed. **This is the most important functional gap in the product.**
- **Job matching is not implemented.** The `matches` table exists; nothing populates it.
- **Mock interview practice is not built.** The `practice_mode` state exists as a placeholder.
- **No CV editing.** A user who wants a change must start over; there is no way to correct a field.
- **Arabic only.** No support for the other first languages common among migrant workers in the region — Urdu, Hindi, Bengali, Tagalog, Amharic — which excludes a large part of the intended audience.
- **Single CV format.** One template, no variants by trade or by employer preference.
- **Sandbox channel.** Users must send a join code before their first message, which is real friction and would not survive contact with a general audience. Production WhatsApp access requires Meta Business verification.

7. Honest assessment

What is genuinely proven: the core loop works. Voice in, professional bilingual CV out, over WhatsApp, with no human in the middle and nothing fabricated. The hardest technical risk — Arabic PDF rendering — is solved properly rather than worked around. Privacy is enforced at the schema level and was verified by testing rather than assumed. The marginal cost is a few cents per CV, which means the economics work for an audience that cannot pay.

What is not yet proven: that it holds up at scale, across dialects, in noisy real-world audio, or in the hands of employers who have never seen it. Those are the next things to test, in that order.